



Create Sales Order Proxy Interface Control Document

VERSION: 2.0

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02/11/2011	Karthik Sreenivasalu	2.0	Initial Draft

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1 Document Details

1.1 Document Location

[CreateSalesOrder_Proxy_ICD.docx](#)

1.2 Reviewers

Name	Title	Date reviewed

1.3 Approvers

<Manager for Consumer/Provider Team>		
Adam Chace	Senior IS Manager – Shared Services	
Preema Pereira	Senior IS Manager – Shared OMS	
Jeff Bridwell	Information Architect	

1.4 Intended audience

This document is intended to be used by but not limited to Staples IS, Enterprise and Application Architects, developers of the consuming applications and/or services.

1.5 ICD Overview

The ICD describes the create sales order proxy request, acknowledgment, and response formats.

- The create sales order proxy can support the creation of a new sales order.
- The create sales order proxy is a public interface that can serve as an integration point to multiple consumer applications.
- The term proxy is synonymous with create sales order proxy.

1.6 References

Canonical Sales Order XSD	ESB Team	CanonicalSalesOrder v2.0.xsd
Canonical Sales Order Acknowledgment XSD	ESB Team	NEED TO ADD LINK

Canonical Sales Order Response XSD	ESB Team	NEED TO ADD LINK
ESB TID Reference	ESB Team	TID02.doc

2 Open Issues

Seq #	Issue Description	Status	Owner	Resolution
1	Submit Order acknowledgement reply message. What should be the average response time?	Open	COM Team	

3 Introduction

3.1 Purpose of the interface

- Multiple applications that wish to create a sales order can integrate with the proxy.
- The proxy decouples the provider applications from consuming applications.
- Consumer and provider applications may either:
 - Send their own application format or
 - Send in the proxy format described below.
- The proxy transforms each consumer message into a canonical sales order format and forwards them to the provider application.
- Consuming applications can consume the proxy format directly or receive the data in a custom format generated by the ESB.

3.2 Scope of the interface

- The proxy transforms and forwards sales order messages supplied by the service consumer application(s).
- The proxy converts messages from the ESB proxy format to the provider specific format on requests.
- The proxy converts messages from the provider specific format to the ESB proxy format on responses.
- The proxy will only apply data level validations and will not perform any business level validations on the request or response messages.
- The proxy may support any QOS from near real time to batch depending on the provider throughput
- The proxy supports persistent messaging.

3.3 Assumptions

Assumption	Impacted Team
Order Create Service would create the order using the order number received from Dotcom.	COM/Dotcom
All user sensitive data (Line Credit card No) needs to be encrypted at source. Only DPS ANs will be accepted by this service.	Dotcom
The Rewards details will be stored and published provided the Rebate service publishes the data	Dotcom
COM will authorize the order amount in full and will not use the 0\$/1\$ authorization details provided by Dotcom.	COM/Dotcom

3.4 Risks

None identified.

3.5 List of Functionalities

Create Sales Order – refer to the Appendix below for samples and message structure.

3.6 Service Category

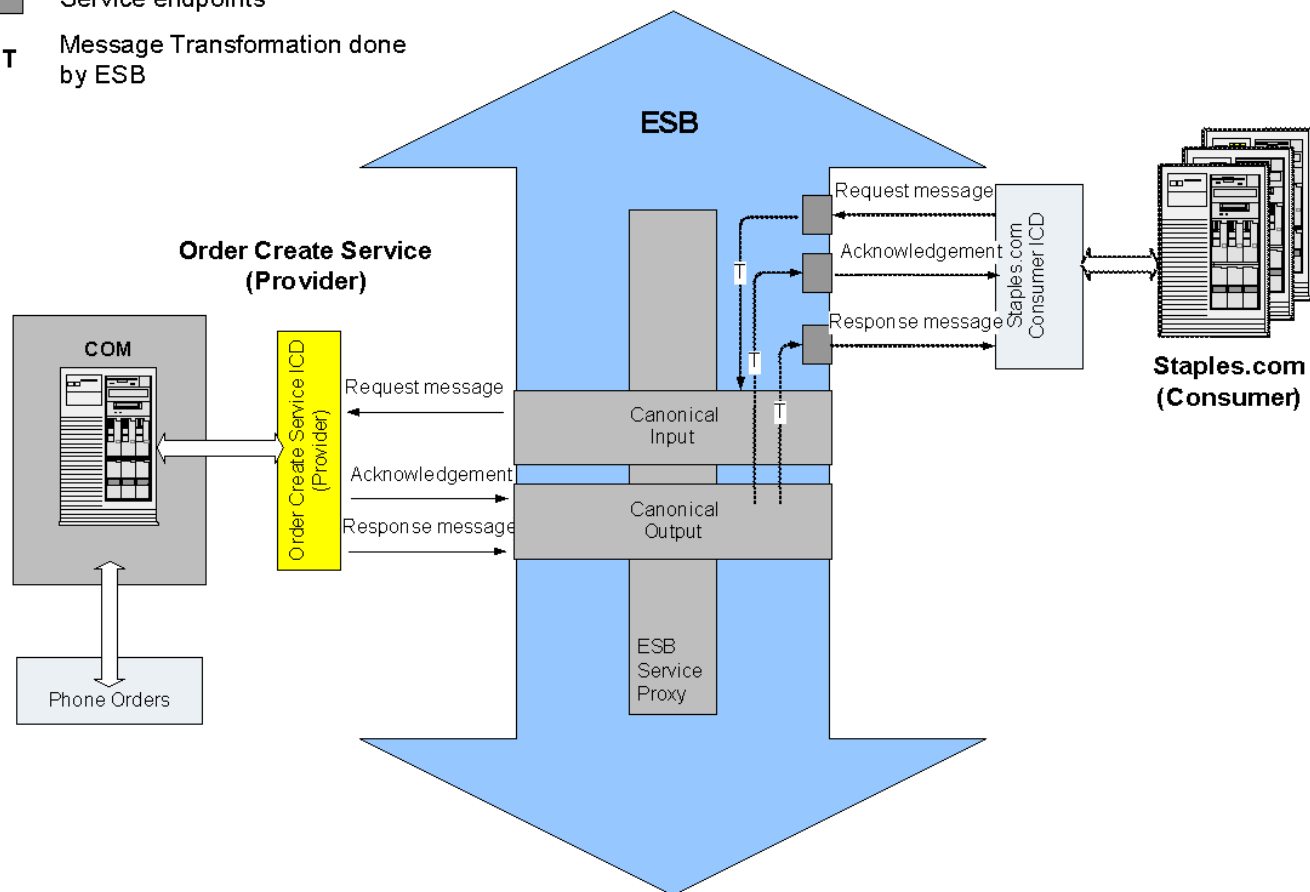
<i>Service category description</i>	
Business process services	☐
Business transaction services	☒
Business function services	☐
Technical function services	☐

3.7 System Context

Order Create Context Diagram

Legend :

- Service endpoints
- T Message Transformation done by ESB



3.8 Service interdependencies

Below are dependencies on service with respect to message types, these service should be available during order processing

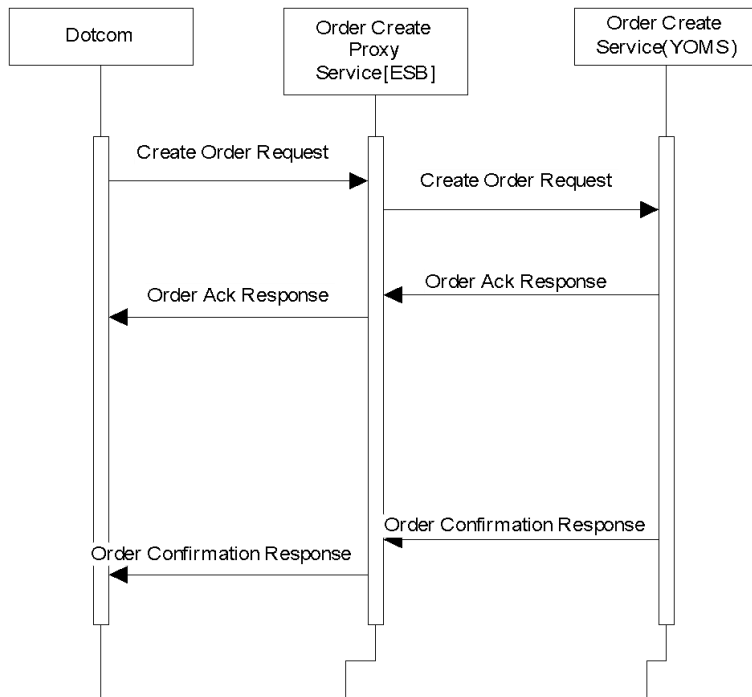
Order Acknowledge Message

- Order Capture System(Dotcom)

Order Confirmation Message

- Enterprise Customer Information Service (ECIS)
- Enterprise Customer Number Assignment Service (ECNAS)
- Coupon Validation and Qualification Service
- Enterprise Tax Service
- Rebate Qualification
- Order Fulfillment Logic Service (OFL)

3.8.1 Sequence Diagrams



3.9 Glossary

4 Functionality Detail

<This section provides detail of each method supported by this interface. The section 4.1 will be repeated once for each method listed in section 3.5 List of Functionalities.>

4.1 Operations

CreateSalesOrder

4.1.1 Abstract

< High level description of the functionality offered by the interface/service >

4.1.2 Functionality Provided

<Provide detail functionality of this operation.>

4.1.3 Exception Behavior

<Provide detail exception processing for this operation. Provide detail of how the business would behave when this operation faces an exception condition.>

4.1.4 Side Effects

<Provide impact and side effects of this operation, if any, on the entire system. For services of type Business Function Service and Technical Function Service, this section will be empty. For services of type Business Process Service and Business Transaction Service, this section must describe the impacts.>

4.1.5 Recommended Usage

Directly consume the proxy XML over a MQ connection.

4.1.6 Invocation type

Asynchronous.

4.1.7 Protocol

XML via MQ.
FlatFiles via MQ.

4.1.8 Request and Response Message

4.1.8.1 Structure

<Describes how logical segments of the message fit together to form a whole>

<For XML/XSD/WSDL structures – Consider an embedded diagram or outline that represents the important segments of the XSD>

4.1.8.2 Semantics

Business Name / Description	Field Name	Path	Data Type	Data Length	Data Format	R/O/C	Repeat	Default Values	Allowed Values	Required Values
Time Entered			String							
Date Entered			String							
Associated SKU			String							
Tax Exempt Mismatch Flag			String							
Customer number the order was placed against on the OMS. ECIS Customer Number as referred to in the PRD.			String							
Sales Order Number	OrderNmb	/SalesOrder/Header/	String							
Catalog code	Code	/SalesOrder/Header/Catalog/	String							
Delivery fee, calculated on website	@Amount	/SalesOrder/Header/ChargeList/Delivery/AmountList/Charge/	String							
Clarify! E-mail language preference. Added for staples.com	EmailLanguage	/SalesOrder/Header/CodeList/	String							
Sales channel id. The channel through which this order was created.e.q Internet, etc	Id	/SalesOrder/Header/CodeList/Channel/	String							
Ship to store	ShipToStore	/SalesOrder/Header/IndicatorList	String							
Indicates whether or not the order originated from an affiliate and contains SKUS to credit to affiliate. Added for staples.com. Y=Affiliate order N=Not affiliate order	AffiliateOrder	/SalesOrder/Header/IndicatorList/	String							
Clarify! E-mail address opt in. Y=Opt in N=Opt out. Added for staples.com	EmailAddressOptIn	/SalesOrder/Header/IndicatorList/	String							
Prepaid order flag	PrepaidOrder	/SalesOrder/Header/IndicatorList/	String							
Business unit number	Nmb	/SalesOrder/Header/OrganizationList/BusinessUnit/	String							
Sales order method	OrderMethod	/SalesOrder/Header/Origin/	String							

Clarify! Session where order originated	SessionId	/SalesOrder/Header/Origin/	String								
Primary e-mail address for a party	EmailId	/SalesOrder/Header/PartyList/Primary/EmailList/Primary/	String								
Person first name	FirstName	/SalesOrder/Header/PartyList/BillTo/NameList/Name	String								
Person last name	LastName	/SalesOrder/Header/PartyList/BillTo/NameList/Name	String								
BillTo Company name	Name	/SalesOrder/Header/PartyList/BillTo/OrganizationList/Company/Name	String								
Standard day phone number extension for a party	Extension	/SalesOrder/Header/PartyList/BillTo/PhoneList/Day	String								
Standard day phone number for a party	Number	/SalesOrder/Header/PartyList/BillTo/PhoneList/Day	String								
Billing Address line 1 where person is located	AddressLine1	/SalesOrder/Header/PartyList/BillTo/PostalAddress/	String								
Billing Address line 2 where person is located	AddressLine2	/SalesOrder/Header/PartyList/BillTo/PostalAddress/	String								
City (within a country) where the person is located	City	/SalesOrder/Header/PartyList/BillTo/PostalAddress/	String								
Postal code for person	PostalCode	/SalesOrder/Header/PartyList/BillTo/PostalAddress/	String								
State (within a country) where the person is located	Name	/SalesOrder/Header/PartyList/BillTo/PostalAddress/State	String								
Customer number which is stored with the customers website profile, based on the customer used on their last online order.	Nmb	/SalesOrder/Header/PartyList/Customer/	String								
Number of employees in person's company	NumberOfEmployees	/SalesOrder/Header/PartyList/Primary/	String								
Standard industrial classification code	Code	/SalesOrder/Header/PartyList/Primary/codeList/StandardIndustrialClassification/	String								
Person first name	FirstName	/SalesOrder/Header/PartyList/ShipTo/NameList/Name	String								
Person last name	LastName	/SalesOrder/Header/PartyList/ShipTo/NameList/Name	String								
Shipping company name	Name	/SalesOrder/Header/PartyList/ShipTo/OrganizationList/Company/Name	String								
Standard day phone number extension for a party	Extension	/SalesOrder/Header/PartyList/ShipTo/PhoneList/Day	String								
Standard day phone number for a party	Number	/SalesOrder/Header/PartyList/ShipTo/PhoneList/Day	String								
Shipping Address line 1 where person is located	AddressLine1	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								

Shipping Address line 2 where person is located	AddressLine2	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								
City (within a country) where the person is located	City	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								
Shipping Address identifier	Id	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								
Shipping Address identifier	Id	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								
Postal code for person	PostalCode	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/	String								
State (within a country) where the person is located	Name	/SalesOrder/Header/PartyList/ShipTo/PostalAddress/State	String								
Customer number entered on the website by the customer. They are intending to place the order against this number.	Nmb	/SalesOrder/Header/PartyList/Web/	String								
For parties using a web site, the number of visits they have made to the site	WebVisitNmb	/SalesOrder/Header/PartyList/Web/	String								
The 4 digits of the payment method that should be displayed	DisplayNmb	/SalesOrder/Header/PaymentmethodList/PaymentMethod/	String								
Payment method number	Nmb	/SalesOrder/Header/PaymentmethodList/PaymentMethod/	String								
Payment amount	@Amount	/SalesOrder/Header/PaymentmethodList/PaymentMethod/AmountList/Payment/	String								
Purchase order identifier (surrogate)	Id	/SalesOrder/Header/PurchaseOrderList/Purchase	String								
Purchase order type	Type	/SalesOrder/Header/PurchaseOrderList/Purchase	String								
Retail Associate Id. Id associated with a person.	Id	/SalesOrder/Header/Retail/Associate/	String								
Retail store location number	Nmb	/SalesOrder/Header/Retail/Store/	String								
Retail Kiosk number	Nmb	/SalesOrder/Header/Retail/Store/	String								
Standard amount for a sub total	@Amount	/SalesOrder/Header/TotalList/Cart/AmountList/SubTotal/	String								
Fee Description	Description	/SalesOrder/Lines/Line/ChargeList/Additional/Charge/	String								
Fee Type	Type	/SalesOrder/Lines/Line/ChargeList/Additional/Charge/	String								
Fee Amount	@Amount	/SalesOrder/Lines/Line/ChargeList/Additional/Charge/AmountList/Charge/	String								
SKU	ItemNmb	/SalesOrder/Lines/Line/CommissionList/Commission/	String								
Publisher number	PublisherNmb	/SalesOrder/Lines/Line/CommissionList/Commission/	String								

Shopper number	ShopperNmb	/SalesOrder/Lines/Line/CommissionList/Commission/	String								
Advertisement number	Nmb	/SalesOrder/Lines/Line/CommissionList/Commission/AdvertisementCampaign	String								
Coupon	Nmb	/SalesOrder/Lines/Line/CouponList/Coupon/	String								
Coupon Value	@Amount	/SalesOrder/Lines/Line/CouponList/Coupon/AmountList/Paid/	String								
Item description	Description	/SalesOrder/Lines/Line/ItemList/Ordered/	String								
Inventory Reservation ID	Nmb	/SalesOrder/Lines/Line/ItemList/Ordered/InventoryReservation/	String								
Inventory Reservation ID Expiration Date	Date	/SalesOrder/Lines/Line/ItemList/Ordered/InventoryReservation/DateList/Expiration/	String								
Standard item price	Amount	/SalesOrder/Lines/Line/ItemList/Ordered/PriceList/Unit/Amount/	String								
Quantity expressed in UnitOfMeasure	Quantity	/SalesOrder/Lines/Line/ItemList/Ordered/QuantityList/Additional/Quantity/	String								
Server IP address. Server information for the described application.	IPAddress	/SalesOrder/TransactionList/Transaction/ApplicationList/Application/ServerList/Server/	String								
This is the total value of the coupon which is the sum of the prorated values	/SalesOrder/Header/CouponList/Coupon/AmountList/Original/	@Amount	String								
Amount that was actually charged to card. We don't have values for this all the time in the request due to prepaid cards.	/SalesOrder/Header/PaymentMethodList/PaymentMethod/AmountList/Payment	@Amount	String								
Reward Amount	/SalesOrder/Header/RebateList/Rebate/Reward/AmountList/Reward/	@Amount	String								
Total tax amount.	/SalesOrder/Header/TaxList/Total/AmountList/Tax	@Amount	String								
The is the sum of all the coupons used.	/SalesOrder/Header/TotalList/Coupons/Amount/	@Amount	String								
This is subtotal as shown on confirm email.	/SalesOrder/Header/TotalList/Merchandise/Amount/	@Amount	String								
Dollar value	/SalesOrder/Lines/Line/CouponList/C	@Amount	String								

	oupon/AmountList/ Paid/										
Unit price amount	/SalesOrder/Lines/ Line/ItemList/Order ed/PriceList/Unit/A mount/	@Amount	String								
Order total amount	/SalesOrder/Lines/ Line/TotalList/Order	@Amount	String								
Coupon Code	/SalesOrder/Heade r/CouponList/Coup on/	Code	String								
Reward Type Code	/SalesOrder/Header/ RebateList/Rebate/R eward/	Code	String								
Code	/SalesOrder/Lines/ Line/CouponList/C oupon/	Code	String								
Shipment carrier code	/SalesOrder/Lines/ Line/ShipmentList/ Shipment/Carrier	Code	String								
Payment method account expiration date.	/SalesOrder/Header/ PaymentMethodList/ PaymentMethod/Dat eList/Expiration/	Date	String								
Rebate Post By/Submit By Date	/SalesOrder/Header/ RebateList/Rebate/D ateList/PostBy/	Date	String								
Rebate Promotion Beginning Date	/SalesOrder/Header/ RebateList/Rebate/Pr omotion/DateList/Be gin/	Date	String								
Rebate Promotion End Date	/SalesOrder/Header/ RebateList/Rebate/Pr omotion/DateList/En d/	Date	String								
Shipment delivery date	/SalesOrder/Lines/ Line/ShipmentList/ Shipment/DateList/ Delivery/	Date	String								
This is a desc such as "\$5 off"	/SalesOrder/Heade r/CouponList/Coup on/	Description	String								
Payment method description	/SalesOrder/Header/ PaymentMethodList/ PaymentMethod/	Description	String								
Rebate Description	/SalesOrder/Header/ RebateList/Rebate/	Description	String								

Reward Description	/SalesOrder/Header/ RebateList/Rebate/R eward/	Description	String								
Description of the coupon.	/SalesOrder/Lines/ Line/CouponList/C oupon/	Description	String								
Item description	/SalesOrder/Lines/ Line/ItemList/Order ed/	Description	String								
The 4 digits of the payment method that should be displayed	/SalesOrder/Heade r/PaymentMethodLi st/PaymentMethod/	DisplayNmb	String								
Primary e-mail address for a party	/SalesOrder/Heade r/PartList/Primary/E mailList/Primary/	EmailId	String								
Express check out. Y=Express N=Standard. Added for staples.com	/SalesOrder/Heade r/IndicatorList/	ExpressCheckOut	String								
staples.com internal customer id	/SalesOrder/Heade r/PartyList/Web/	Id	String								
Shipment identifier	/SalesOrder/Lines/ Line/ShipmentList/ Shipment/	Id	String								
Phone number (excluding area code, country code and extension). If the ESB receives a merged number (where the parts are not separated), map the whole number here.	/SalesOrder/Heade r/PartyList/BillTo/Fa xList/Day/	Nmb	String								
Phone number (excluding area code, country code and extension). If the ESB receives a merged number (where the parts are not separated), map the whole number here.	/SalesOrder/Heade r/PartyList/ShiptoTo /FaxList/Day/	Nmb	String								
Purchase Order number	/SalesOrder/Heade r/PurchaseOrderLis t/Purchase	Nmb	String								
Rebate Promotion Number	/SalesOrder/Header/ RebateList/Rebate/Pr omotion/	Nmb	String								
Item number	/SalesOrder/Lines/ Line/ItemList/Order ed/	Nmb	String								
Shipment number	/SalesOrder/Lines/ Line/ShipmentList/ Shipment/	Nmb	String								

Print Form Id (Y=Display N=Do Not Display)	/SalesOrder/Header/ RebateList/Rebate/In dicatorList/	PrintFormId	String								
Quantity expressed in UnitOfMeasure	/SalesOrder/Lines/ Line/ItemList/Order ed/QuantityList/Ad ditional/Quantity/	Quantity	String								
Sequence number of this coupon (in relation to other coupons applied)	/SalesOrder/Heade r/CouponList/Coup on/	SequenceNmb	String								
Sequence number of this coupon (in relation to other coupons applied)	/SalesOrder/Lines/ Line/CouponList/C oupon/	SequenceNmb	String								
Serial Number Required (Y=Required N=Not Required)	/SalesOrder/Header/ RebateList/Rebate/In dicatorList/	SerialNmbRequired	String								
Ship to store	/SalesOrder/Heade r/IndicatorList	ShipToStore	String								
Submit Online (Y=Easy Rebate N=Mail-In)	/SalesOrder/Header/ RebateList/Rebate/In dicatorList/	SubmittableOnline	String								
Item suffix	/SalesOrder/Lines/ Line/ItemList/Order ed/	suffix	String								
Payment method type	/SalesOrder/Header/ PaymentMethodList/ PaymentMethod/	Type	String								
Reward Type Description	/SalesOrder/Header/ RebateList/Rebate/R eward/	Type	String								

4.1.8.3 Examples

<Provide at least one example. It can be a required elements only message first. Include multiple examples of core cases as development proceeds>

4.1.8.4 Links

[CanonicalSalesOrder_v2.0.xsd](#)

4.1.9 Quality of Service (QoS) Requirements

4.1.9.1 Security Requirements

4.1.9.1.1 Authentication Requirements

<Are any special authentication requirements required?>

4.1.9.1.2 Encryption Requirements

<Is encryption required? If so, describe the specific encryption type needed>

4.1.9.2 Performance (Response Times)

<What is the current or desired SLA? If consumers plan to consume a providers service as part of the scope of the project, what are their end-to-end SLA requirements?>

Response Time Measurement	Value	Description
Maximum Response Time (MRT)		MRT at normal load
Maximum Response Time Service Level Agreement		Percentage of time the MRT will be met at normal load (should be >99%)
Average Response Time (ART)		ART at normal load
Average Response Time Service Level Agreement		Percentage of time the ART will be met at normal load (should be >95%)

4.1.9.3 Transaction Volume (Throughput)

<What are expected volumes? Peak day, peak hour, average day, average hour, min/max/average message sizes>

Transaction Volume	Value	Description
Active days each week		List the days of week where transaction volumes are expected
Quality of service (QOS)	Batch Event Near real time	Describe the volume pattern for the interface. Describe the QOS the application wants. Note that the initial QOS ask may shift during the design and performance testing phases. Batch – SLAs for interface data are > 15 minutes Event – SLAs for interfaces > 6 seconds and < 15 minutes Near real time – SLAs for interfaces are < 6 seconds
Peak number of transactions per hour (across all consumers/providers)		
Peak number of transactions per day (across all consumers/providers)		
Average number of transactions per hour (across all consumers/providers)		Based on historical data from service (if available)
Average number of transactions per day (across all consumers/providers)		Based on historical data from service (if available)
<Line of business> Peak transactions per hour		Transactions/hour based on Line of Business (SBD: Base of 50K orders per hour. Base of 75K shipments per hour)

<Line of business> Peak transactions per day		Transactions/hour based on Line of Business (SBD: Base of 120K orders per day. Base of 180K shipment per day)
--	--	--

4.1.9.4 Availability

The sales order proxy is required to be available 24x7, 365 days a year with the exception of planned maintenance windows.

4.1.9.5 Reliability (Mean time between failures)

<From IEEE – “the ability of a system or component to perform its required functions under stated conditions for a specified period of time”>

4.1.9.6 Timeouts

<What is the expected behavior in case of time outs?>

Service Request Type	Response Type
Return Lookup	TBD
Return History	TBD
Return Authorization	TBD